March 5, 2026

Bruce Stallsmith and Kara Million
Editors, Special Issue on Parasites in Freshwater Ecosystems
Journal of Freshwater Ecology

Dear Drs. Stallsmith and Million,

Enclosed, please find a manuscript titled, "How's the weather in there? Hurricanes drive short-term declines in the abundance of direct life cycle parasites of riverine fish," which has been invited to the special issue of the Journal of Freshwater Ecology on parasites in freshwater ecosystems.

Anthropogenic climate change is accelerating the frequency and intensity of extreme weather events like hurricanes. These events have known effects on human infectious disease dynamics, but little is known about how infectious disease agents in wildlife populations are impacted by these storms. Our study investigates the impacts of hurricanes on riverine fish parasite populations in the Pearl River between 1963 and 2005. This long-term dataset was generated using parasitological dissections of over 1000 museum specimens, and it intersects with 24 hurricanes and importantly incorporates a Before-After-Control-Impact (BACI) design, where the 'impact' is hurricane presence; the two main effects (before-after, control-impact) interact to create four categories (before hurricane, after hurricane, before control, after control). This novel application of the BACI design allows us to systematically quantify hurricane effects on parasites of riverine fish.

We discovered that direct life cycle parasites decline following hurricane events on short (3-month) and long (6-month) timescales. This study complements existing literature on the effects of climate change on parasitism, suggesting that direct life cycle parasites are susceptible to short-term and severe extreme weather events like hurricanes, while complex life cycle parasites are more resilient to such effects.

This exciting step forward offers a novel analytical approach for mapping extreme weather events using publicly available storm data and expands our understanding of how hurricanes impact ecosystems and their inhabitants.

My co-authors and I are eager to hear your thoughts on this work. Thank you in advance for considering this manuscript for publication in the Journal of Freshwater Ecology.

Thank you,

Connor Whalen

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